Contents

1 Introduction

Over the last decade tremendous progress has been made in the field of artificial intelligence (AI). From self-driving cars to protein folding: machine learning (ML) has shown itself to be inevitable for future technology. But not only has it transformed the industry world, it has also become an integral part of scientific research. The processing and analysis of huge amounts of data, which are typical for experiments in high energy or astrophysics, pose major challenges to traditional methods. With the rise of AI, we can tackle these tasks in a reasonable way. But what about theoretical physics? The questions there might not involve a lot of data, but could be even more difficult to answer. They are however of great importance for the future, because they can give us a deep understanding of nature, which is very often the basis for the invention of new technologies.

One such topic, which has attracted a lot of interest, is the holographic principle. It appeared first in the context of string theory and states, that quantum gravity can be encoded on the boundary of its spacetime. It can therefore be thought of as a hologram of the information stored in lower dimensional space. It turns out, that this actually defines a non-gravitational theory on the boundary, which should be equivalent to the higher-dimensional bulk theory. Two physical theories without any apparent connection are somehow ultimately the same. This is also called the gauge-gravity duality, because the boundary system is often found to be a gauge theory. The gravity theory is related to it in a special way: whenever one theory is in the strong coupling regime, the other is weekly coupled. This opens up the possibility to analyze a system at energy scales, at which perturbation theory would usually break down. To do that we simply have to find and use its holographic dual. If this relationship actually holds, then we have a powerful tool at hand to access information about gravitational systems or even quantum gravity itself. Black holes are examples for such systems, which can be explored via holography. As a matter of fact, the holographic principle was inspired from a result about the entropy of black holes. It was discovered that the maximal information contained in a black hole scales with its boundary surface [7]. On a more practical side, the duality could enable us to investigate strong coupling phenomena of the gauge theories like holographic

QCD. Although this theory is not the same as the physical QCD with $SU(3)$ color gauge group, it could still give us insights about aspects common to QCDs. An important concept is for example the confinement of quarks, which is still puzzling to this day. An understanding of its underlying mechanism is tightly intertwined to the question of stability of protons. With the holographic principle we might be able to tackle these difficult tasks. Unfortunately, it is not clear how to find the precise mapping between the two descriptions of the same physics.

There exist various versions and formulations of the principle and the most popular duality is arguably the AdS/CFT correspondence. Its general statement is that any conformal field theory (CFT) defined a conformally flat spacetime of d dimensions is equivalent a quantum gravity theory in a bulk spacetime, which is asymptotically $AdS_{d+1} \times C$, with C being some compact manifold. Although this correspondence has been analyzed in great detail over the last two decades, it still remains unproven. The equivalence between gauge theory and gravity is only conjectured. Could machine learning perhaps help us to gain a deeper understanding of the correspondence?

Since the appearance of neural networks many architectures have been introduced, which have produced impressive results. Of particular interest are the generative models, which can learn a (sometimes even unkown) probability distribution and create samples from it. Such networks are for example variational autoencoders (VAEs), Boltzmann machines, general adversarial networks (GANs) or normalizing flows. The ability to sample efficiently from a learned distribution is especially important, when the original probability density cannot be computed due to its complexity. This is for example the case for quantum field theories, where the partition function can only be calculated for Gaussian distributions. In this situation flow networks can help us. By using this network architecture to approximate the probability distribution, we can not only sample, but also can the learned density be directly accessed, i.e. it is tractable. This is one of the reasons, why normalizing flows will be our network of choice.

In order to apply this network to holography, we have to make use of a peculiar fact about the interior of the bulk space. As mentioned above, the CFT is defined on the AdS boundary. When we move from the boundary into the bulk space along the radial (or holographic) direction, which is perpendicular to the boundary surface, we can find another CFT. This new theory turns out to be a renormalized version of the one on the boundary. The flow into the bulk interior corresponds to an RG-flow of the CFT and the radial coordinate can be identified with the inverse RG-scale. If we can set up up a neural network, which can perform an RG-flow on a field theory, then we might find an emergent geometric structure in the network corresponding to a hyperbolic space.

In the following we will first study the AdS/CFT correspondence in more detail to gain an better understanding of the physical framework. After that, the flow networks are reviewed alongside with features to be used later. We are then ready to build an RG-flow neural network and apply it to a disrete QFT on a lattice. The training of various networks is discussed and methods to extract geometrical information from the flow are presented. At last we will mention some applications for the trained flow and show possible extensions of the model.

2 AdS/CFT Correspondence

A first concrete example of the holographic principle was given by Maldacena in 1997 [22], when he conjectured the equivalence between a $\mathcal{N} = 4$ supersymmetric $SU(N)$ Yang-Mills theory with conformal symmetry and string theory on $\mathrm{AdS}_5 \times S^5$. This was motivated by the fact, that by taking specific limits of type IIB string theory in a background curved by N coincident D3-branes one can get either one of the two theories (see [10] for a detailed review). The equivalence is already apparent at the level of symmetry. Since the two theories are conjectured to be the same, their symmetry groups have to coincide. In general, the conformal field theory lives in a d dimensional flat spacetime and the gravity theory is defined in a $d + 1$ dimensional hyperbolic background. A hyperbolic space is usually defined as a hypersurface of a higher dimensional flat space. This means that an AdS_5 space inherits the Lorentz isometry $SO(4, 2)$ from its embedding, which has 15 generators. The symmetries of a CFT on the other hand are generated by the usual 10 Lorentz generators corresponding to translations, rotations and boosts with additional 5 conformal symmetries, which include four special conformal and one scale transformation. These 15 generators satisfy the same algebra as the isometry group of the AdS space. In fact, $SO(4, 2)$ is also called the conformal group. On the gravity side we further have a $SO(6)$ symmetry from the compactification on the sphere S^5. This corresponds to $\mathcal{N} = 4$ supersymmetry of the field theory, which has symmetry group $SU(4)$, because the two groups are isomorphicto each other: $SO(6) \cong SU(4)$.

To see that the CFT actually lives on the boundary of the bulk space, we look at AdS_{d+1}-metric global coordinates [6]

$$\mathrm{d}s^2 = \frac{L^2}{\cos^2 \rho}(-\,\mathrm{d}\tau^2 + \mathrm{d}\rho^2 + \sin^2 \rho \,\mathrm{d}\Omega^2) \ , \tag{2.1}$$

with $-\pi < \tau \leq \pi$, $0 \leq \rho < \frac{\pi}{2}$ and $\sum_{i=1}^{d} \Omega_i^2 = 1$. We observe that the boundary is the hypersurface at $\rho = \frac{\pi}{2}$ and has the topology of $\mathbb{R} \times S^{d-1}$, which is conformal to d-dimensional Minkowski space. Therefore, a conformally invariant QFT can be thought of as living on the conformal boundary of AdS_{d+1}. It contains the same amount of information as the gravity theory. Or in other words, the bulk theory can be holographically projected and encoded onto the boundary surface. With a change of coordinates and a conformal rescaling we can be bring the metric into the simple form

$$\mathrm{d}s^2 = \frac{L^2}{z^2}(\mathrm{d}z^2 + \eta_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j) \ , \tag{2.2}$$

which is referred to as Poincaré coordinates and covers half of AdS_{d+1}. This is most common and convenient metric of AdS space. Its boundary lies at $z = 0$ with Minkowski metric η_{ij}.

For easier calculation and implementation we will choose to use the Euclidean version of AdS space, the hyperbolic space $\mathbb{H}^{d+1}$. We will however this refer to this space as AdS, since a Euclidean metric can be easily obtained from the Minkowski version by Wick rotation to imaginary time. With this, the boundary metric becomes $\eta_{ij} = \delta_{ij}$. Bulk indices will be denoted by greek letters μ, ν and boundary indices by latin letters i, j. We will use $x, y \in \mathbb{R}^d$ as coordinates of the boundary with $x = (x^0, ..., x^d)$ and $z \in \mathbb{R}$ will usually be the holographic radial direction.

To analyze the correspondence we would like to find some observables, which we can compute in both theories. Comparing results can then give us a first check, whether the holographic principle is realized at all. A simple quantity to calculate is the two-point function or correlator, which will be our focus.

2.1 Conformal two-point function

Due to the large group of Poincaré and conformal transformations, the general form of the one, two and three-point functions are completely determined by symmetry considerations up to constant factors. Suppose we have two complex fields $\psi_1(x), \psi_2(y)$ with conformal dimensions Δ_1, Δ_2 on the boundary. Because of translation invariance, we already know that the correlator between the two fields has be a function of the distance

$$\langle \psi_1(x)\psi_2^*(y)\rangle = f(|x - y|) \ . \tag{2.3}$$

In a CFT we are allowed to perform scaling transformations $x \to \lambda x$ without changing the observables. A field will transform under scaling like

$$\psi_i(\lambda x) = \lambda^{\Delta_i}\psi_i(x) \ . \tag{2.4}$$

We can then restrict the form of the two-point function. The constraint reads

$$\begin{aligned}\langle \lambda^{\Delta_1}\psi_1(\lambda x)\lambda^{\Delta_2}\psi_2^*(\lambda y)\rangle &= \lambda^{\Delta_1+\Delta_2} f(\lambda(x-y)) \\ &\stackrel{!}{=} f(x-y)\end{aligned} \quad , \tag{2.5}$$

which is satisfied, when f decays according to a power-law in the distance $|x-y|$:

$$f(|x-y|) \sim \frac{1}{|x-y|^{\Delta_1+\Delta_2}} \tag{2.6}$$

One can further find a relation between the conformal dimensions Δ_1 and Δ_2. A symmetry, that plays a very important role in CFTs, is the invariance under inversions

$$x \to \frac{x}{|x|^2} \ . \tag{2.7}$$

After performing this transformation on the correlator and imposing invariance, we find the condition:

$$\begin{aligned}\left\langle \frac{1}{x^{2\Delta_1}y^{2\Delta_2}}\psi_1(-\frac{1}{x})\psi_2^*(-\frac{1}{y})\right\rangle &= \frac{1}{x^{2\Delta_1}y^{2\Delta_2}}\frac{c}{(-\frac{1}{x}+\frac{1}{y})^{\Delta_1+\Delta_2}} \\ &\stackrel{!}{=} \frac{c}{|x-y|^{\Delta_1+\Delta_2}}\end{aligned} \ . \tag{2.8}$$

To fulfill this constraint, we have to set $\Delta_1 = \Delta_2$. This means, that the correlator between two fields is only non-zero, if their dimensions are the same. So in general, the two-point function in a conformal field theory will always be an algebraic decay in the form

$$\langle \psi(x)\psi^*(y)\rangle \sim \frac{1}{|x-y|^{2\Delta}} \tag{2.9}$$

This gives us a practical way to check, if a theory is not or cannot be described by a CFT. When a physical is in a highly disordered state, the correlation function will usually be an exponential decay with some finite correlation length. A power-law implies then, that the correlation length has (formally) diverged.

2.2 Boundary values as sources

A first precise formulation of the correspondence was given by Witten [28], where he showed how to calculate the two-point function of a CFT using the bulk gravity action. We will follow his derivation of the boundary correlator. Assume that for operators $\mathcal{O}$ in the CFT there exist a term in the field theory action, that couples them to an external source J. In general, the interaction between them has the form

$$\int \mathrm{d}x\, \mathcal{O}(x) J(x) \tag{2.10}$$

The usual procedure for a QFT is to construct a partition function $Z[J]$ and calculate the n-point functions as functional derivates of $Z[J]$ with respect to the external source at $J = 0$. In the following, we will express the bulk action in terms of the boundary values of the classical bulk fields.

As the bulk theory, consider the case of a free complex scalar field $\phi(x, z)$ in a general background with action

$$S_{\text{bulk}}[\phi, g_{\mu\nu}] = \int \mathrm{d}^d x\, \mathrm{d}z\, \sqrt{g}\left[g^{\mu\nu}\partial_\mu \phi^* \partial_\nu \phi + m^2 |\phi|^2 + R\right] \ . \tag{2.11}$$

To find the classical equation of motion for ϕ, we integrate by parts. After fixing the geometry to be AdS_{d+1}, we get for the ϕ-dependent terms

$$S_{\text{bulk}}[\phi] = -\int \mathrm{d}^d x\, \mathrm{d}z\, \sqrt{g}\phi^*(-\Box_g + m^2)\phi + \int \mathrm{d}^d x\, \mathrm{d}z\, \partial_\mu(\sqrt{g} g^{\mu\nu}\phi^*\partial_\nu\phi) \tag{2.12}$$

with the Laplacian

$$\begin{aligned} \Box_g &= \frac{1}{\sqrt{g}}\partial_\mu(\sqrt{g} g^{\mu\nu}\partial_\nu) \\ &= \frac{z^2}{L^2}(\partial_z^2 + (1-d)\frac{1}{z}\partial_z + \partial_x^2) \ . \end{aligned} \tag{2.13}$$

By imposing $\delta S = 0$, we find the equation of motion from the first term in equation (2.12). The second term has no contribution, because it can be written as a surface integral by invoking Stokes' theorem. The e.o.m. then reads

$$z^2\partial_z^2\phi - (d-1)z\partial_z\phi + (z^2\partial_x^2 - L^2 m^2)\phi = 0 \ , \tag{2.14}$$

where the general solution will be a convolution of a boundary configuration and the Green's function K satisfying

$$(-\Box_g + m^2)K(x, z) = \delta(x, z) \ . \tag{2.15}$$

To solve this, we look for a solution of the homogeneous equation, which has a singularity at one point on the boundary. This corresponds to the behavior we would expect for a δ-function. Since we have translation invariance in x direction, the Green's function should only be a function of z and we take as an ansatz

$$K(z) = z^\alpha \ . \tag{2.16}$$

By changing coordinates using the inversion map $x \to \frac{x}{z^2+|x|^2}$, $K(z)$ becomes

$$K(z) = \frac{z^\alpha}{(z^2+|x|^2)^\alpha} \ . \tag{2.17}$$

This function has a singularity at $x = 0$ for $z \to 0$, which means it behaves like a δ-function on the boundary. We can now express our solution as the convolution of the Green's function with a source:

$$\phi(x,z) \sim \int \mathrm{d}y\, K(x-y,z)\phi_0(y) \ , \tag{2.18}$$

where $\phi_0(x) \equiv \phi(x, z=0)$.

We can use this expression to write the action as a functional of the boundary values of $\phi(x,z)$. As mentioned above, the second term in the action (2.12) can be expressed as a surface integral with induced metric h_{ij} and normal vector n^μ as

$$\int \mathrm{d}^d x\, \sqrt{h}\phi^* n^\mu \partial_\mu \phi = \int \mathrm{d}^d x\, \frac{1}{z^{d-1}}\phi^* \partial_z \phi \ . \tag{2.19}$$

The derivative with respect to z can be estimated to lowest order in z from the classical solution (2.18) as

$$\partial_z \phi \sim z^{d-1} \int \mathrm{d}y\, \frac{\phi_0(y)}{|x-y|^{2\alpha}} \ . \tag{2.20}$$

With this, the action on the boundary will be of the form

$$S_{\text{bulk}}[\phi_0] = \int \mathrm{d}x\, \mathrm{d}y\, \frac{\phi_0^*(x)\phi_0(y)}{|x-y|^{2\alpha}} \ . \tag{2.21}$$

From this expression, the two-point function for the CFT can be calculated.

Assume, that we have the following coupling of a complex boundary field $\psi(x)$ with an external source $\phi_0(x)$:

$$-\int \mathrm{d}x\, [\psi(x)\phi_0^*(x) + \psi^*(x)\phi_0(x)] \ . \tag{2.22}$$

Since the AdS/CFT correspondence states, that the boundary and bulk theories are the equivalent, their partition functions should contain the same information, i.e.

$$Z_{\text{CFT}} = Z_{\text{bulk}} \approx e^{-S_{\text{bulk}}[\phi_0]} \ . \tag{2.23}$$

In the saddle point approximation of the gravity theory, the classical bulk action plays the role of the generating functional for the field theory and we can compute the two-point function for ψ by taking a functional derivatives of Z_{bulk} w.r.t. the sources ϕ_0:

$$\begin{aligned}\langle \psi^*(x)\psi(y)\rangle &= \frac{\delta}{\delta\phi_0(x)}\frac{\delta}{\delta\phi_0^*(y)} Z_{\text{bulk}}[\phi_0] \\ &\sim \frac{1}{|x-y|^{2\alpha}} \ .\end{aligned} \tag{2.24}$$

Thus, we have found a polynomial decay of the correlation function, which is consistent with the result from the CFT, and can identify the exponent with the conformal dimension, $\alpha = \Delta$. We have obtained way to calculate any n-point function of the QFT without precise knowledge of its action or dynamics, since the information is provided by the bulk theory. One should note, that this is the weakest form of the AdS/CFT correspondence, because it is only valid in the saddle point approximation of the bulk theory, e.g. the supergravity limit of string theory. In the case of the original correspondence involving SUSY Yang-Mills theory with gauge group $SU(N)$, this implies the large N limit $N \to \infty$. However, the equivalence is believed to hold even in the quantum regime and on all energy scales.

2.3 Holographic renormalization

When calculating observables of a QFT, one usually has to do them carefully in a renormalized fashion, as quantities can easily diverge due to the presence of arbitrary small scales or infinite volumes in the definition of the bare action. A potential divergence is for example the integration $\int \mathrm{d}^d x$ over the boundary at $z = 0$, since the area is infinite. Before we perform any calculations, we have to regularize the theory by introducing a cutoff to the length or energy scale, e.g. area integral goes over a surface at some $z_0 \neq 0$. We can send $z \to 0$ after the calculation to remove the cutoff. This transformation is part of the isometry $(x, z) \to (\lambda x, \lambda z)$ of AdS space, which is dual to the dilation symmetry of CFTs. For the boundary theory this corresponds to a coarse-graining, which induces an RG-flow.

As an example let's consider again a massive field $\phi(x, z)$ in the bulk. A typical correlation function between two points in the bulk at $z = z_0$ will decay exponentially with length of paths connecting them [13] as

$$\langle \phi(x)\phi^*(y) \rangle_{z=z_0} \sim \sum_{\text{paths}(x,y)} e^{-m\mathcal{D}(x,y)} \ . \tag{2.25}$$

This expression will be dominated by the geodesic between the points (x, z_0) and (y, z_0) for $m \gg L^{-1}$, such that

$$\langle \phi(x)\phi^*(y) \rangle_{z=z_0} \sim e^{-m\mathcal{D}_{\text{geo}}(x,y)} \ . \tag{2.26}$$

The geodesic in AdS space will be of logarithmic dependence on $|x - y|$ and is given by

$$\mathcal{D}_{\text{geo}}(x,y) = L \ln \left(\frac{\left(|x-y| + \sqrt{|x-y|^2 + z_0^2} \right)^2}{z_0^2} \right) \ . \tag{2.27}$$

For large distances $|x - y| \gg z_0$, this can be approximated as

$$\mathcal{D}_{\text{geo}}(x,y) \approx L \ln |x-y| \ . \tag{2.28}$$

We now see, that the two-point function in a hyperbolic space will be approximately a power-law given by

$$\langle \phi(x)\phi^*(y) \rangle_{z=z_0} \sim \frac{1}{|x-y|^{2mL}} \ . \tag{2.29}$$

This is the familiar form, that we get for a CFT with identification $\Delta = mL$. However, this functional dependence is only approximate and the precise behavior at short scales is modified and is not polynomial. This implies, that the long distance physics of the bulk theory is related to the UV of the boundary. This is called the IR-UV relation or duality, which was first discussed in [26]. The theory, that we find at z_0 is, due to the effective UV-cutoff induced by the IR-cutoff in the bulk, the renormalized CFT at energy scale

$$\Lambda_0 = \frac{1}{z_0} \ . \tag{2.30}$$

This implies a relation of the renormalization group flow of the CFT with the radial flow of the gravity theory, see figure 2.1. The information of the interior space enclosed by the dotted line is encoded on its surface area. The Bekenstein bound for the entropy of this spacetime volume is satisfied.

In fact, one can make this relationship more concrete. But as with the calculation of the

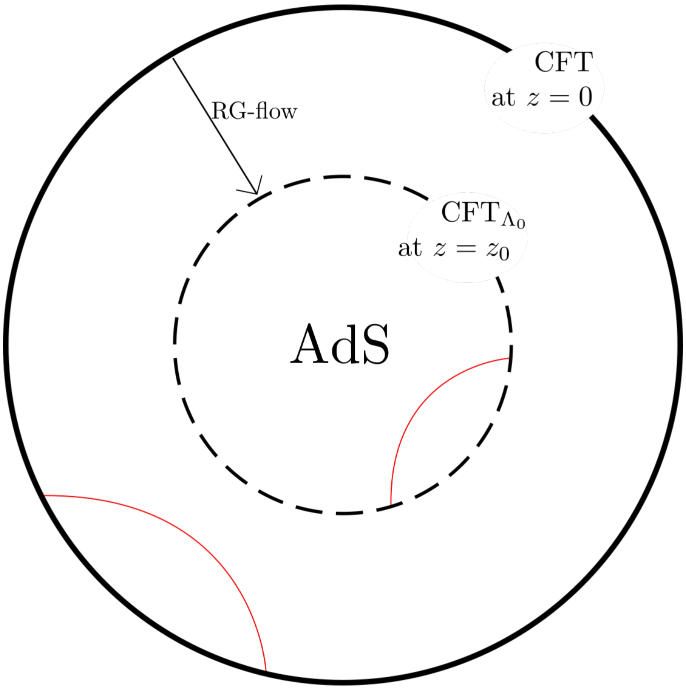


Figure 2.1: The flow in the radial direction z of AdS space is equivalent to an RG-flow of the CFT. The blue lines represent bulk geodesics.

two-point function, a precise computation is only possible in the saddle point approximation.

In a Wilsonian approach we would integrate out the UV degrees of freedom in the CFT, which should correspond to an integration of the long wavelength modes in the bulk [17]. In the classical limit, we can then define an inner correspondence of the CFT at z_0 and the interior volume of AdS space by integrating out bulk fields in the excluded area for $z < z_0$ [4].

A more precise mapping can be achieved when we go into the Hamilton-Jacobi formulation of the gravity theory, where the first order flow equations can be cast into the form of Callan-Symanzik equation [8]. In a semiclassical treatment the field values $\phi(x, z)$ can be interpreted as the running coupling constants of the CFT. The Hamilton equations of motion of the bulk fields are then equivalent to the renormalization group flow equations of the CFT [3] and we can map

$$z\frac{\mathrm{d}}{\mathrm{d}z}\phi = \frac{\delta\mathcal{H}}{\delta\pi} \leftrightarrow \Lambda\frac{\mathrm{d}}{\mathrm{d}\Lambda}g = \beta(g)\ , \tag{2.31}$$

where $\mathcal{H}$ is the Hamiltonian, π the momentum variable conjugate to ϕ, Λ the energy scale for the RG-flow of some coupling constant g in the CFT. The metric of the bulk is therefore a metric of the space of RG-flow and the value of the field ϕ at different radial positions is the running coupling constant. However, such a mapping is only valid in the semiclassical limit and also depends on the renormalization scheme. That means that we have to take precautions when setting up a coarse-graining procedure for a CFT, if we want the RG-flow space to be described a metric. Because there exist various renormalization schemes, not all RG-flow spaces will exhibit a geometric structure even if there exist a dual gravity description for the CFT.

3 Normalizing Flows

An important goal in statistical modeling is to sample from a desired target probability distribution $q_{\text{target}}(x)$. Unfortunately, very often we are not able to do so directly, because the probability density of interest is not tractable. This is usually due to high dimensional integrals, which appear in the calculation of the normalization constant. To circumvent this problem, we can make use of Monte Carlo methods like the Metropolis-Hastings algorithm, where knowing a function proportional to q_{target} is sufficient. However, Monte Carlo sampling can be computationally quite expensive. A more efficient approach is to utilize generative ML models. One of the first network structures with the ability to capture a probability density are Boltzmann machines. The binary nodes of this type of memory model, a so-called Hopfield network, behave exactly the same way like spins in an Ising model. This means, if we choose the energy of the Ising model as the loss function, we are effectively searching for an equilibrium state during training. The configuration of the units will then follow the Boltzmann distribution $p \sim e^{-E}$. Although samples from this density can be easily obtained by taking random configurations of the nodes, the underlying distribution function is not accessible. This drawback is also present in other more efficient ML models like autoencoders or GANs.

A solution is provided by normalizing flows [25]. These neural networks can evaluate the learned distribution exactly, because it is obtained by the application of a series of invertible transformations on a simple base distribution, typically taken to be Gaussian. The flow network effectively performs a change of variables on normal distributions. Since it is invertible, the reverse flow map can also be used. Training the network then equates to finding a set of maps, which transform a complicated probability density into the product of normal distributions, i.e. the network normalizes the target distribution. A detailed review of normalizing flows is [23], which we will largely follow.

We will present basic concepts for flow networks in the case of complex variables. This is a bit unusual, because neural networks are usually build with real variables and complex numbers are defined as 2D vectors. But there has been great effort recently to implement complex numbers natively into the standard ML-libraries.

In this chapter we will denote the data vector by $x = (x^1, ..., x^N)^T \in \mathbb{C}^N$ and the (latent) variable of the base distribution by $z = (z^1, ..., z^N)^T \in \mathbb{C}^N$, which is common in the ML literature.

3.1 Tractable probability density

In the ideal case, the network transformation G^{-1} maps the input data x, which follows the target distribution $q_{\text{target}}(x)$, to the latent random variables $G^{-1}(x) = z \sim p_{\text{base}}(z)$. Conversely, we can also sample from the base distribution $p_{\text{base}}(z)$ by applying the inverse map on z to obtain new data $x = G(z)$. The two distributions are related by a change of variables with the Jacobian matrix

$$J_G(z) = \begin{pmatrix} \frac{\partial G^1}{\partial z^1} & \cdots & \frac{\partial G^1}{\partial z^N} \\ \vdots & \ddots & \vdots \\ \frac{\partial G^N}{\partial z^1} & \cdots & \frac{\partial G^N}{\partial z^N} \end{pmatrix} . \tag{3.1}$$

Because we are using complex numbers and $\mathbb{C} \cong \mathbb{R}^2$, the vector x actually represents $2N$ independent variables. So when we change from x to z, we have to account for contributions coming from x and also its conjugate $x^\dagger$. If our flow treats the real and imaginary part of x and z in the same way, i.e. they transform under the same map, then $\det J^*_{G^{-1}}(x) = \det J_{G^{-1}}(x^\dagger)$ and the target density can then be rewritten as

$$\begin{aligned} p_{\text{flow}}(x) &= p_{\text{base}}(G^{-1}(x)\left|\det \frac{\partial G^{-1}}{\partial x}\right|\left|\det \frac{\partial G^{-1}}{\partial x^\dagger}\right| \\ &= p_{\text{base}}(z)\left|\det \frac{\partial G}{\partial z}\right|^{-1}\left|\det \frac{\partial G}{\partial z^\dagger}\right|^{-1} \\ &= p_{\text{base}}(z)|\det J_G(z)|^{-2} , \end{aligned} \tag{3.2}$$

where the switch from G^{-1} to G is allowed by the inverse function theorem. When we choose a suitable base distribution, we can to take almost any density, even a very restricted one like the uniform distribution. In fact, if $q_{\text{target}}(x)$ and $p_{\text{base}}(z)$ are well-behaved, there always exist a diffeomorphism to transform one to the other (see [23] for a proof).

The extension to multiple bijective maps G_k is straightforward and done by simple composition of the transformations, see figure 3.1. The full map of the flow consisting of K

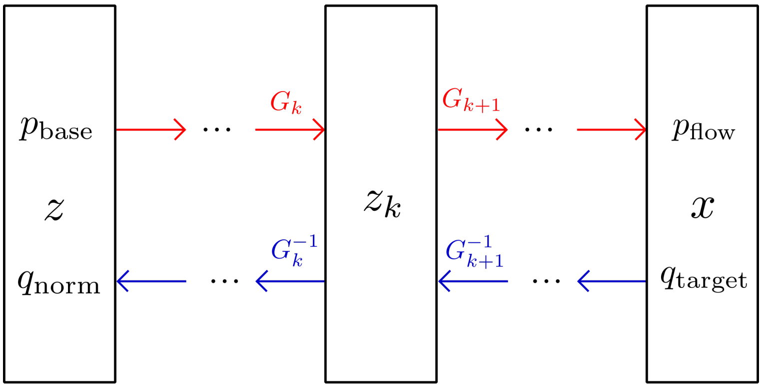


Figure 3.1: Flow network consisting of multiple bijective maps

invertible layers is then given by

$$G = G_K \circ \cdots \circ G_1 \tag{3.3}$$

with inverse transformation

$$G^{-1} = G_1^{-1} \circ \cdots \circ G_K^{-1} \ . \tag{3.4}$$

The Jacobian determinant for G is still easily accessible as a product of each layer:

$$\begin{aligned} \det J_G(z) &= \det J_{G_K}(G_{K-1}(z_{K-2})) \cdots \det J_{G_1}(z) \\ &= \prod_{k=1}^{K} \det J_{G_k}(z_{k-1}) \ , \end{aligned} \tag{3.5}$$

where we defined $z_k = G_k(z_{k-1})$ and identified $z_0 \equiv z$, $z_K \equiv x$. By collecting the Jacobian at each layer during a pass through the flow network, one can easily obtain the determinant of the full transformation. Very often, we will not directly handle the determinants, but instead calculate the logarithm of them. This is because in practice one usually works with log probabilities, as they are additive and therefore easier to manipulate. This is also favorable from the computational point of view, because determinants can easily blow up in value for high dimensional data. Calculating the logarithm of Jacobians is more stable. For the expression (3.5) we get

$$\log|\det J_G(z)| = \sum_{k=1}^{K} \log|\det J_{G_k}(z_{k-1})| \ , \tag{3.6}$$

which is now a summation over the layer contributions instead of a product.

3.2 Unitary flows

The process of designing the transformation G_k is usually guided by the premises to have tractable Jacobians and inverse map. Ideally, the log determinants are easy to compute and the inverse map has a simple form. But the only necessary requirement is the existence of the inverse or a non-zero Jacobian. One could implement G_k as entire neural networks, which is the case for flows like RNVPs. In the simplest case, G_k is just a linear layer. By constraining the weight matrix to be orthogonal or unitary, we get a trivial Jacobian and $\log|\det J| = 0$. The computation of the inverse for unitary matrices also requires no special solving algorithm. We will discuss two ways to transform a weight matrix $\tilde{W}$ into a unitary layer U.

Assuming we have a upper triangluar matrix with zeros on the diagonal of the form

$$\tilde{W} = \begin{pmatrix} 0 & w_{1,2} & \dots & w_{1,n} \\ 0 & 0 & \ddots & \vdots \\ \vdots & & \ddots & w_{n-1,n} \\ 0 & 0 & \dots & 0 \end{pmatrix} . \tag{3.7}$$

$\tilde{W}$ can be made into a skew-hermitian (or anti-hermitian) matrix with

$$W = \tilde{W} - \tilde{W}^\dagger . \tag{3.8}$$

Since the space of skew-hermitian matrices with zero trace form the Lie algebra of the special unitary group $SU(n)$, we can apply the exponential map on W and get a unitary layer U for the flow:

$$U = e^W . \tag{3.9}$$

Another possibility to construct a unitary matrix from the anti-hermitian W is to use the so-called Cayley transform, which is defined as

$$U = (I - W)^{-1}(I + W) , \tag{3.10}$$

where I is the identity matrix. The inverse transformation is given by

$$W = (U - I)(U + I)^{-1} . \tag{3.11}$$

One should note, that we cannot generate any element of $SU(n)$ by this transformation, since we are excluding certain matrices, for example those which are unitary but not skew-hermitian. Both options, the Cayley and exponential map, take $\mathcal{O}(n^3)$ time to compute, which will become inefficient for large matrices.

3.3 Kullback-Leibler divergence

In order to train a flow network for an inference or modeling task, an appropriate choice for the loss function has to be made. The most common choice is the relative entropy or Kullback-Leibler divergence. It serves as a measure for how much two distributions differ from each other. It is important to know that the divergence is not symmetric in its arguments. For flow networks we can distinguish between the forward KL divergence and the reversed one.

The forward version is defined as

$$\begin{aligned} D_{\mathrm{KL}}(q_{\mathrm{target}}(x) \parallel p_{\mathrm{flow}}(x)) &= \sum_x q_{\mathrm{target}}(x) \log\left(\frac{q_{\mathrm{target}}(x)}{p_{\mathrm{flow}}(x)}\right) \\ &= \mathbb{E}_{x\sim q_{\mathrm{target}}}\left[\log q_{\mathrm{target}} - \log p_{\mathrm{flow}}\right] \\ &= \mathbb{E}_{x\sim q_{\mathrm{target}}}\left[\log q_{\mathrm{target}} - \log|\det J_{G^{-1}}| - \log p_{\mathrm{base}}\right] \ , \end{aligned} \tag{3.12}$$

where the expectation value over the target distribution is carried out as an averaging over the training set. The KL-divergence is zero, if the two distributions are equal. The divergence can be interpreted as the additional entropy, if one uses the distribution p_{flow} instead of the target q_{target} to sample x. The expectation over q_{target} will give us a constant term, such that the loss function can be defined as

$$\mathcal{L} = \mathbb{E}_{x\sim q_{\mathrm{target}}}\left[\log|\det J_G| - \log p_{\mathrm{base}}\right] \tag{3.13}$$

It is worth noting that in this approach we do not need to calculate $G(z)$, although it is of course required for sampling. In the holographic network, which will be presented in the next chapter, we do not have a training set at hand. The evaluation of the expectation value is therefore not possible. However, we can use the relative entropy function with reversed arguments, such that the loss includes an expectation value over the network distribution, i.e. the averaging over a sampled data set.

The reversed KL divergence is more suitable for our purposes and is defined as

$$\begin{aligned} D_{\mathrm{KL}}(p_{\mathrm{flow}}(x) \parallel q_{\mathrm{target}}(x)) &= \mathbb{E}_{x\sim p_{\mathrm{flow}}}\left[\log p_{\mathrm{flow}} - \log q_{\mathrm{target}}\right] \\ &= \mathbb{E}_z\left[\log p_{\mathrm{base}} - \log|\det J_G| - \log q_{\mathrm{target}}\right] \ . \end{aligned} \tag{3.14}$$

For the evaluation of the expectation value of q_{target}, a function proportional to the target distribution $\tilde{p}_{\text{target}} \sim q_{\text{target}}$ is sufficient, because the normalization constant will give a constant term due to the logarithm. Our loss function can then be chosen to be

$$\mathcal{L} = \mathbb{E}_z \left[\log p_{\text{base}} - \log |\det J_G| - \log \tilde{p}_{\text{target}} \right] \tag{3.15}$$

Training of the flow under reversed KL-divergence means that we are trying to find an approximation of q_{target} by fitting a function p_{flow} to the target distribution. This process is called probability density distillation.

There is an interesting connection between the training procedures under the two versions of the KL-divergence. Let us first define the probability distribution $q_{\text{norm}}(z)$, which is the density induced by the network transformation via normalizing $q_{\text{target}} \xrightarrow{G^{-1}} q_{\text{norm}}$. It can be found by changing variables from x to z for $q_{\text{target}}(x)$:

$$q_{\text{norm}}(z) = q_{\text{target}}(x) |\det J_G|^2 \ . \tag{3.16}$$

A forward KL-divergence can be obtained from the reversed version by using the above transformation:

$$\begin{aligned} D_{\text{KL}}(p_{\text{flow}} \parallel q_{\text{target}}) &= \mathbb{E}_z \left[\log p_{\text{base}}(z) - 2 \log |\det J_G(z)| - \log q_{\text{target}}(G(z)) \right] \\ &= \mathbb{E}_z \left[\log p_{\text{base}}(z) - \log q_{\text{norm}}(z) \right] \\ &= D_{\text{KL}}(p_{\text{base}}(z) \parallel q_{\text{norm}}(z)) \ . \end{aligned} \tag{3.17}$$

We see that fitting the distribution $p_{\text{flow}}(x)$ to the target $q_{\text{target}}(x)$ under the reversed KL-divergence is equivalent to fitting the induced density $q_{\text{norm}}(z)$ to the base distribution under the forward divergence.

3.4 Reparametrization trick

The typical choice for the base density is a normal distribution. In the simplest case each variable $z^\alpha \in \mathbb{C}$ is drawn from $\mathcal{N}_{\mathbb{C}}(0, 1)$ with density

$$p(z^\alpha) = \frac{1}{\pi} e^{-|z^\alpha|^2} \ . \tag{3.18}$$

Effectively, the real and imaginary part of z^α are sampled from real normal distributions with variance $\frac{1}{2}$, i.e. $\text{Re}(z^\alpha), \text{Im}(z^\alpha) \sim \mathcal{N}(0, \frac{1}{2})$. All variables z^α are i.i.d. and have no correlation among them:

$$\mathcal{C}(z^\alpha, z^\beta) = \mathbb{E}[z^\alpha, z^\beta] = 0 \quad \forall z^\alpha, z^\beta \ . \tag{3.19}$$

This is puts a lot of restriction on the information, that can be stored in the base distribution. A less biased choice is a multivariate complex Gaussian

$$p_{\text{base}}(z) = \frac{1}{\det(\pi\Sigma)} \exp\left[-\frac{1}{2} z^\dagger \Sigma^{-1} z\right] \ , \tag{3.20}$$

where $\Sigma = \mathbb{E}[zz^\dagger]$ is the covariance matrix. Its off-diagonal components contain the correlation between different random variables z^α. Because this distribution has zero mean ($\mathbb{E}[z] = 0$) and zero pseudo-covariance ($\mathbb{E}[zz^T] = 0$), it is circularly-symmetric, i.e. z is evenly distributed across all directions in the complex plane.

The matrix Σ can be made learnable via the reparametrization trick. Since the evaluation of the loss function involve the expectation value over the base distribution, it is difficult to parameters contained in p_{base}. The idea is to implement the multivariate Gaussian as a linear layer. We observe, that we can obtain $z \sim p_{\text{base}}(z)$ from a transformation of i.i.d. variables $\epsilon \sim \mathcal{N}_{\mathbb{C}}(0,1)$ via $z = L\epsilon$. The lower triangluar matrix L can be calculated from the Cholesky decomposition of the covariance and satisfies $\Sigma = LL^\dagger$. We can make a quick check by changing variable for the base distribution to get

$$p_{\text{base}}(\epsilon) = \frac{1}{\pi^N} \exp\left[-\frac{1}{2} \epsilon^\dagger \epsilon\right] \ , \tag{3.21}$$

where the covariance matrix is the identity and all correlations between different random variables have been removed.

4 Holographic RG with Flow Networks

When modeling a boundary theory and its holographic bulk with a neural network, the first step is to decide which physical quantity will be represented by the network nodes. One possibility is to use the Hamilton formulation of the gravity dual and identify the units of a sequential net with the values of the bulk field and its conjugate momentum [15]. The forward direction of the network will then represent the radial coordinate z of the bulk space and the weight matrices will be constraint to produce discrete Hamilton equations of motion. To train the model a set of observables of the boundary QFT has to be mapped to the values of the network on the input layer at $z = 0$, as the input layer contain the boundary values of the bulk fields. This has been demonstrated for a holographic QCD theory [16]. The disadvantage of this approach is that the neural network is defined on the phase space of the fields and will therefore not directly model a holographic space, i.e. it is not a disretization of AdS space. The geometric information is contained in the derivatives, which are encoded in the weight constraints. The phase space representation also restricts the size of the sequential net, i.e. the input dimension is fixed to two in the case of one scalar bulk field. Moreover, sampling will be quite difficult or inefficient, as we have no access to the probability distribution of the CFT.

Another option comes from exploiting the fact, that the visible units of a trained restricted Boltzmann machine (RBM) obey the distribution $p \sim e^{-E}$, which can be used to model a QFT, when the energy function is identified with the classical limit of the gravity action [14]. The connections between the layers of the RBM have to be carefully defined by choosing a consistent discretization scheme for the field derivatives, that appear in the action. Further constraints on the network units are also needed to ensure the existence of a saddle approximation for the gravity theory. The RBM can then learn information about the bulk space geometry, because metric components appear in the weights. The network size can be easily scaled, because it is defined in discrete real space.

While this generative model gives us the ability of sampling, the probability distribution is unfortunately not tractable. To circumvent this problem we will use a normalizing flow, which is much more efficient and versatile because of its invertibility and tracetable probability density. But unlike before, we will not model the bulk fields and then infer CFT observables from the boundary values, but we will try to map the CFT degrees of freedom directly to a bulk field. To achieve that, the network will be set up as a renormalization procedure for the boundary QFT. Since we will learn the probability distribution via the use of the reversed KL divergence, there is no need to prepare a training data set. The hyperbolic geometry will be an emergent feature, that will appear after successful training.

4.1 Network architecture

We design the holographic flow with a structure known from tensor networks, the multi-scale entanglement renormalization ansatz (MERA). It was proposed as a method to find the ground state of a critical quantum system [27]. When a wave function is fed to the network, degrees of freedom of a certain length scale will be separated at each layer, where the layer depth corresponds to said scale. Therefore MERA performs a renormalization on the input data and each layer is an RG step. Such a neural network was first implemented in [21], where it was applied to spin states. It was later generalized [19] to other data sets, which are more common in the machine learning community, like CelebA, CIFAR-10 or MNIST. The basis of our discussion and analysis will be [18], where the neural RG was applied to a holographic setup. Our network will be similiar to those, namely it will have the same MERA architecture, but has modifications with different choices of transformations.

The flow has the MERA structure shown in figure 4.1. The depth of the network can be interpreted as the holographic direction. The deeper we go into the interior, the lower the energy scale $\Lambda = z^{-1}$ of the renormalized CFT becomes. Similiar to CNNs the input data will be processed in patches of a specific kernel size $\sqrt{n} \times \sqrt{n}$. On these n variables a set of transformations is applied consisting of a scaling, linear map and an activation function. There are two types of linear transformations: the disentanglers E and the decimators C. The map E serves to reduce the correlation between variables, while C performs a coarse-graining in real space. After a decimator transformation only $\frac{n}{4}$ variables will processed in the next layer. The other three forth of the degrees of freedom are taken to be the latent or bulk variables of the base distribution. As standard kernel size we will choose 2×2 and work with this in the following. An RG-step consists of two layers: one with only disentanglers and the other with only decimators. For two-dimensional data we can distribute E and C in an overlapping manner.

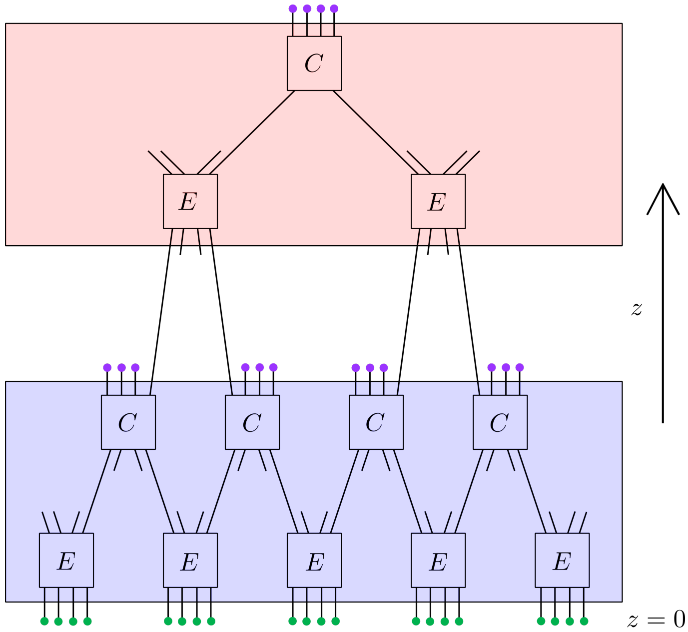


Figure 4.1: General architecture of the holographic flow. The colored boxes are RG-layers at different energy scales with disentanglers E and decimators C. The violet dots are bulk variables and the green dots are CFT variables.

We will use a physics motivated notation, when discussing the network: the boundary variable is ψ, the latent variable will be refered to as bulk field ϕ and the base density will be called the bulk distribution p_{bulk}. The training will happen in two stages. In stage I we will sample the bulk variables from i.i.d. normal distributions and train the network using the generative direction G. This is the standard approach in probability density distillation with normalizing flows. After training the network weights in G will be fixed and we will relax the bulk distribution to be a multivariate Gaussian. We will then try to capture the residual correlation of the bulk variables by training the covariance matrix via the reparametrization trick. The off-diagonal components of Σ will then contain the information about the hyperbolic geometry.

Exact holographic mapping An analytic version of the network above was given in [24] and [20]. It shares a similiar structure as MERA, but with the difference that the RG-steps are only composed of one layer instead of two. In particular, the disentanglers have been omitted. It was shown, that by choosing the linear map to be a unitary matrix, one can coarse-grain a boundary Hamiltonian. The map actually performs a basis change in Hilbert space on the operators. Assuming n incoming operators, the unitary map separates the degrees of freedoms into UV and IR modes. When going from the boundary into the bulk interior, the IR degrees of freedom are fed into the next layer of transformation, while the UV modes are interpreted as bulk operators. This corresponds to a UV-cutoff of the CFT. An explicit matrix can be given, for which the Hamiltonian will split into three terms. One of them is simply a rescaling of the original Hamiltonian and can be identified as the IR mode, since it is the result of a renormalization. The other two terms are then the UV mode and an interaction term. Consider a system on a lattice with an action of the form

$$S = \sum_x \psi_x^* \hat{V}_x \psi_x \ , \tag{4.1}$$

where $\hat{V}_x$ is some interaction operator. A coarse-graining can be easily achieved by grouping together two neighboring spinors at sites $2x$ and $2x-1$ and defining the new degrees of freedom to be proportional to the sum. A rotation like

$$\begin{pmatrix} \psi_x^{IR} \\ \psi_x^{UV} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \psi_{2x-1} \\ \psi_{2x} \end{pmatrix} \tag{4.2}$$

can map ψ_x into renormalized modes ψ_x^{IR} and bulk modes ψ^{UV}. If we rewrite the action in terms of ψ_x^{IR} and ψ^{UV}, the term, which dependents only on ψ_x^{IR}, will be of the same form as the original action except being rescaled with half the number of operators:

$$S_{renorm} = \frac{1}{2} \sum_x \psi_x^{*IR} \hat{V}_x \psi_x^{IR} \ . \tag{4.3}$$

Thus the boundary theory has been renormalized. We can then perform the transformation (4.2) iteratively at each layer of the network on the IR modes ψ_x^{IR} coming from the previous layer. With the precise mapping of boundary degrees of freedom to bulk operators one has obtained an exact holographic mapping (EHM).

Hyperbolic geometry The MERA architecture actually exhibits a geometric structure, which is suitable for holography. Let us look at the geodesic distance between two sites on the boundary, i.e. the shortest path connecting them in the bulk(see figure 4.2). We observe, that the geodesic has to decay logarithmically with the boundary distance as

$$\mathcal{D}_{\text{MERA}}(x, y) = \alpha \log |x - y| \tag{4.4}$$

with some proportionality factor α. This is consistent with a hyperbolic space. To check the ability of the network to model a CFT, we analyze the two-point function. Let's denote a tensor on the boundary site position x by ψ_x. The correlation between two sites at x and y in general decays exponentially in terms of the geodesic distance [11]

$$\langle \psi_x \psi_y^* \rangle \sim e^{-m\mathcal{D}_{\text{MERA}}(x,y)} \ , \tag{4.5}$$

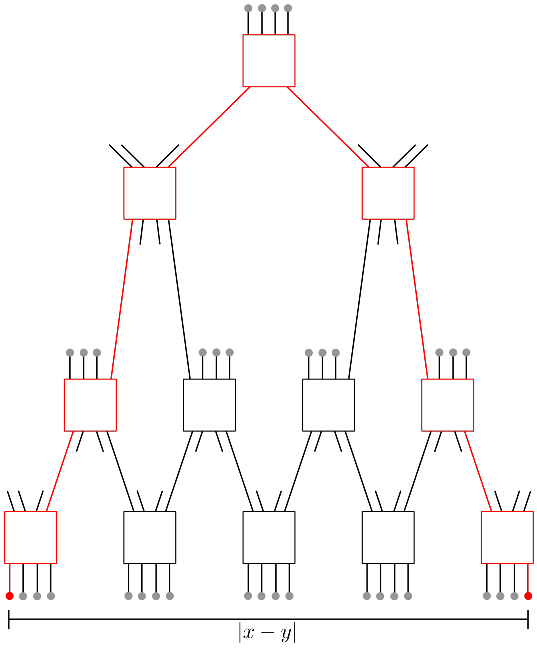


Figure 4.2: The red geodesic is the path with the least number of links connecting the two red CFT sites. The distance $|x-y|$ on the boundary is the separation in terms of site numbers.

into which we can plug in the expression from before. This leads to a polynomial decay of the correlation as one would expect of a system in critical phase:

$$\langle \psi_x \psi_y^* \rangle \sim e^{-m\alpha \log |x-y|} \sim \frac{1}{|x-y|^{-m\alpha}} \ . \tag{4.6}$$

As it turns out, the tensor network also satisfies the area law for entropy. It seems, that MERA is a good candidate for a discrete hyperbolic space in a holographic setup. In order for this AdS/MERA correspondence to be consistent, one should identify the lattice spacing with the AdS radius L by matching geodesic lengths [5]. This means, that the network can only capture physics on scales larger than L and should be regarded as a classical spacetime.

4.2 Holographic system

4.2.1 Discretized theory

Let us elaborate on the physical theory, which we want to model with the network. We will work with bosonic fields on a square lattice with N sites in total. To avoid irregular behavior at the edges, we impose periodic boundary conditions. We work in Poincaré coordinates, where the boundary coordinates will be denoted by $x, y \in \mathbb{R}^2$ and the radial direction by $z \in \mathbb{R}$.

Boundary CFT We choose a complex scalar field ψ living two-dimensional real space, whose dynamics is governed by the action

$$S_{\mathrm{CFT}}[\psi] = -k \sum_{\langle x,y \rangle} \psi_x^* \psi_y + \sum_x (\mu |\psi_x|^2 + \lambda |\psi_x|^4) \;, \tag{4.7}$$

where the sum $\sum_{\langle x,y \rangle}$ is to be taken over nearest neighbors and $J, \mu, \lambda \in \mathbb{R}$ are global parameters. The associated probability distribution is given by

$$q_{\mathrm{CFT}}(\psi) = \frac{1}{Z_{\mathrm{CFT}}} e^{-S_{\mathrm{CFT}}[\psi]} \;. \tag{4.8}$$

If one wants to make connection to Witten's decription of holography, one can add a further term, which couples the field ψ to the boundary value of a bulk field $\phi_0 \equiv \phi(z = 0)$. Such a source term with minimal coupling has the form

$$S_{\mathrm{src}}[\psi] = -\frac{1}{2} \sum_x (\psi_x^* \phi_{0x} + \phi_{0x}^* \psi_x) \;. \tag{4.9}$$

In the following we will work with the case of vanishing external source $\phi \to 0$.

For the parameters we choose $k = 5$, $\mu = -180$, $\lambda = 25$, which creates a mexican hat potential for ψ with $U(1)$ symmetry and a reflection symmetry $\mathbb{Z}_2$, see figure 4.3. During training the field $\psi = \rho e^{i\theta}$ will condensate into the minima and is constrained to have a approximate constant amplitude $|\psi| \approx 2$. If the temperature T is sufficiently low, we can ignore the fluctuations perpendicular to the θ direction. Then only one degree of freedom θ survives and the boundary theory becomes an XY-model with action

$$S_{\mathrm{CFT}} \approx -k\rho^2 \sum_{\langle x,y \rangle} \cos(\theta_x - \theta_y) \tag{4.10}$$

at an effective temperature $T = \frac{1}{k\rho^2}$.

Low temperature limit It is important to mention, that this theory is in general not conformally invariant. While the Mermin-Wagner theorem prohibits any spontaneous symmetry breaking in $d \leq 2$, the system can still experience a topological phase transition. That means, it has a critical phase at low temperatures, where it exhibits a quasi-long-

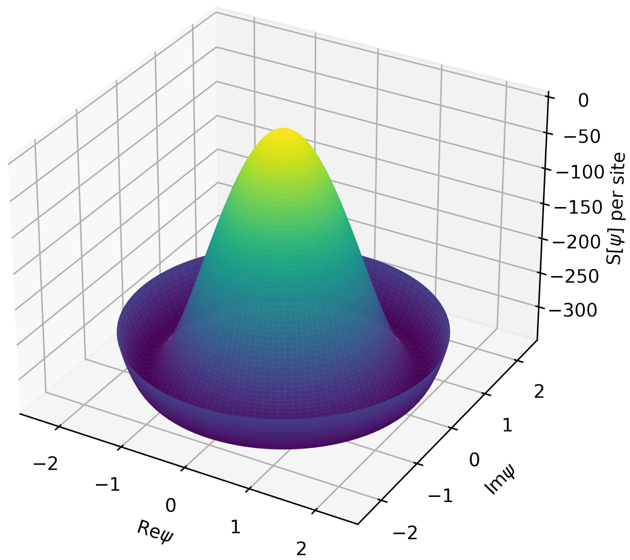


Figure 4.3: Mexican hat potential in the complex plane with minima at $\rho = \sqrt{\frac{37}{10}}$ and $S_{min} = -\frac{1369}{4}$

range interaction and can thus be described by a CFT. There exist a certain temperature T_{KT}, below which the system can form bound pairs of vortices and goes into an ordered phase. The correlation function will obey a power law instead of an exponential decay, which implies an infinite correlation length. This indicates the existence of conformal symmetry. To check this, we will look for an expression of the two-point correlator in the critical phase. Since for low T the fluctuations of θ will be small, we expand the action (4.10) up to quadratic order:

$$-k\rho^2 \sum_{\langle x,y\rangle} \cos(\theta_x - \theta_y) = -2Nk\rho^2 + \frac{1}{2}k\rho^2 \sum_{\langle x,y\rangle} (\theta_x - \theta_y)^2 + \dots \, . \tag{4.11}$$

Within this harmonic approximation we expect small variations of the fields, i.e. $\theta(x) \ll 1$, such that the subtraction can be seen as a derivative by taking the continuum limit. So we arrive at a free field theory for $\theta(x)$ with

$$S_{\mathrm{CFT}} \approx \frac{k\rho^2}{2} \int \mathrm{d}^2 x \, (\nabla\theta(x))^2 \, . \tag{4.12}$$

The two-point function can be found from the Green's function of the Laplace equation in d dimensions and reads

$$\begin{aligned}\langle\theta(x)\theta(y)\rangle &= -\frac{1}{k\rho^2}\frac{|x-y|^{2-d}}{(2-d)S_d} + \text{const.} \\ &\xrightarrow{d=2} -\frac{1}{2\pi k\rho^2}\ln\frac{|x-y|}{a} \ ,\end{aligned} \tag{4.13}$$

where S_d is the solid angle and a the lattice spacing. The two-point correlation of our original field ψ can then approximated as

$$\begin{aligned}\left\langle\psi_x\psi_y^*\right\rangle &\sim \left\langle e^{\mathrm{i}(\theta_x-\theta_y)}\right\rangle \\ &= e^{-\frac{1}{2}\langle(\theta_x-\theta_y)^2\rangle} \\ &\sim \frac{1}{|x-y|^{\frac{1}{2\pi k\rho^2}}} \ ,\end{aligned} \tag{4.14}$$

where the second line is valid, since we have a Gaussian distribution due to (4.12). The same estimation can be done for the real and imaginary part and for those we get

$$\begin{aligned}&\operatorname{Im}\left\langle\psi_x\psi_y^*\right\rangle \approx 0 \\ &\operatorname{Re}\left\langle\psi_x\psi_y^*\right\rangle \sim \left|\left\langle\psi_x\psi_y^*\right\rangle\right| \sim \frac{1}{|x-y|^{\frac{1}{2\pi k\rho^2}}} \ .\end{aligned} \tag{4.15}$$

This power-law is consistent with the result from a CFT. For the XY-model it means, that there exist a ordered phase at low T and the system can undergo a phase transition, the so-called Kosterlitz-Thousless transistion. Our parameter choice leads to a temperature of $T \approx 0.05$, which is much lower than the estimated crititical temperature at $T_{\mathrm{KT}} \approx 0.9$. The emergence of a dual gravity theory in the network can therefore be expected.

Bulk gravity The physical system on the gravity side can potentially contain a large set of different fields. We assume, that on large scales the general behavior can be described effectively by a massive complex scalar field $\phi(x,z)$ with action

$$S_{\text{bulk}}[\phi] = \int \mathrm{d}^d x\,\mathrm{d}z\,\sqrt{g}\left[g^{\mu\nu}\partial_\mu\phi^*\partial_\nu\phi + m^2|\phi|^2\right] \tag{4.16}$$

without any interaction of higher order. Since this action is quadratic in ϕ, we can discretize it by a general ansatz

$$S_{\text{bulk}}[\phi] \approx \sum_{\alpha,\beta}\phi_\alpha^\dagger\Sigma_{\alpha\beta}^{-1}\phi_\beta = \phi^\dagger\Sigma^{-1}\phi \ , \tag{4.17}$$

where indices go over the bulk coordinates, i.e. $\alpha = (x, z)$. This leads to a circularly-symmteric complex normal distribution of N random variables

$$p_{\text{bulk}}(\phi) = \frac{1}{\det(\pi\Sigma)} e^{-\phi^\dagger \Sigma^{-1} \phi} \ . \tag{4.18}$$

The covariance matrix Σ contains the geometric information about the bulk space, so we have to make it learnable by the reparametrization trick. The base distribution will then be chosen to be a complex normal distribution with $\Sigma = 1$, i.e.

$$\tilde{p}_{\text{bulk}}(\varphi) = \frac{1}{\pi^N} e^{-\varphi^\dagger \varphi} \ , \tag{4.19}$$

and the bulk field can be obtained by

$$\phi = L\varphi \ , \tag{4.20}$$

where L can be calculated from the Cholesky decomposition of the covariance matrix $\Sigma = LL^\dagger$.

4.2.2 Holographic distances

Since we know that geodesics in AdS space between two points (x, y) at the same radial position z_0 go as $\mathcal{D}_{\text{geo}} \sim \ln|x - y|$, we can check for a hyperbolic geometry by analyzing the functional dependence of geodesics in our network. The distance in the bulk space can the inferred from the correlator on large scales. If the bulk field is massive, the two-point function will decay exponentially with the distance between two points in the bulk as

$$\langle \phi_\alpha \phi_\beta^* \rangle \sim \exp\left[-\frac{\mathcal{D}_{\text{holo}}(\phi_\alpha, \phi_\beta)}{\xi} \right] \ , \tag{4.21}$$

This can be used as a definition of the geodesic distance, because the correlations can be extracted from the covariance matrix $\Sigma_{\alpha\beta} = \langle \phi_\alpha \phi_\beta \rangle$ or from a batch of sampled bulk states as $\langle \phi_\alpha \phi_\beta^* \rangle = \mathbb{E}_\phi[\phi_\alpha \phi_\beta^*]$. So we define

$$\mathcal{D}_{\text{geo}}(\phi_\alpha, \phi_\beta^*) = -\xi \ln \frac{|\langle \phi_\alpha \phi_\beta^* \rangle|}{C_0} \ , \tag{4.22}$$

where C_0 is a normalization constant, we have to determine. The offset C_0 is chosen as the average value of the diagonal entries of the covariance. We expect that Σ will have roughly the same value for every variable, and that the diagonal values are much larger than the off-diagonals. This would be the case of a very massive field, which was our assumption for (4.21). The constant can then be estimated to be

$$C_0 = \frac{1}{N}\sum_\alpha |\langle\phi_\alpha\phi_\alpha^*\rangle| \approx |\langle\phi_\alpha\phi_\alpha^*\rangle| \ . \tag{4.23}$$

This choice will make the holographic distance between bulk variables at the same position to zero:

$$\mathcal{D}_{\text{geo}}(\phi_\alpha,\phi_\alpha) = -\xi\ln\frac{|\langle\phi_\alpha\phi_\alpha^*\rangle|}{C_0} \approx 0 \tag{4.24}$$

for any ϕ_α.

Mutual information The definition of distance as (4.22) has the disadvantage, that it depends on the particular pair of field variables and we might overlook correlations. A maybe more reliable option is the mutual information, defined as

$$I(\phi_\alpha:\phi_\beta) \equiv H(\phi_\alpha) + H(\phi_\beta) - H(\phi_\alpha,\phi_\beta) \ , \tag{4.25}$$

where H is the entropy. We can use I to bound the correlator from above. The mutual information can also be expressed by the Kulback-Leibler divergence as

$$I(\phi_\alpha:\phi_\beta) = D_{\text{KL}}(p(\phi_\alpha,\phi_\beta)) \parallel p(\phi_\alpha)p(\phi_\beta)) \ . \tag{4.26}$$

By using Pinsker's inequality for the KL-divergence we can get a bound for the two-point function:

$$\begin{aligned}
I(\phi_\alpha:\phi_\beta) &\geq \frac{1}{2}\left(\sum_{\phi_\alpha,\phi_\beta}|p(\phi_\alpha,\phi_\beta) - p(\phi_\alpha)p(\phi_\beta)|\right)^2 \\
&\geq \frac{1}{2}\left(\sum_{\phi_\alpha,\phi_\beta}|(p(\phi_\alpha,\phi_\beta) - p(\phi_\alpha)p(\phi_\beta))|\left|\frac{\phi_\alpha\phi_\beta}{|\phi_\alpha\phi_\beta|}\right|\right)^2 \\
&= \frac{1}{2}\frac{\left|\langle\phi_\alpha\phi_\beta^*\rangle\right|^2}{|\phi_\alpha|^2|\phi_\beta|^2} \ .
\end{aligned} \tag{4.27}$$

Let us calulcate the mutual information for a bivariate complex Gaussian, to get a formula for the information between two variables. This will be the effective distribution, which contains the correlation between two bulk site and it can be obtained by marginalization. The joint entropy for a multivariate normal distribution of N variables can be calculated as

$$
\begin{aligned}
H(\phi) &= -\int \mathrm{d}\phi \, \mathrm{d}\phi^\star \, p \ln p \\
&= \ln|\det(\pi\Sigma)| + \langle \phi^\dagger \Sigma^{-1} \phi \rangle \\
&= \ln|\det(\pi\Sigma)| + \langle \mathrm{tr}\big(\phi\phi^\dagger \Sigma^{-1}\big) \rangle \\
&= N \ln(\pi) + \ln|\det(\Sigma)| + N \; .
\end{aligned} \tag{4.28}
$$

Plugging this expression into equation (4.25) gives us

$$
I(\phi_\alpha : \phi_\beta) = \ln\left(1 - \frac{\langle \phi_\alpha \phi_\beta^* \rangle \langle \phi_\beta \phi_\alpha^* \rangle}{\langle \phi_\alpha \phi_\alpha^* \rangle \langle \phi_\beta \phi_\beta^* \rangle}\right) \; , \tag{4.29}
$$

which gives us an alternative way to calculate the geodesic distance:

$$
\mathcal{D}_{\mathrm{geo}}(\phi_\alpha, \phi_\beta) = -\xi \ln \frac{I(\phi_\alpha : \phi_\beta)}{I_0} \; , \tag{4.30}
$$

where I_0 is a normalization constant corresponding to the case of maximal entanglement.

While the bulk sample will have the same shape as the boundary data on the lattice, the bulk variables are actually spread in the radial direction z, i.e. they belong to different length scales of the RG-flow. The assignment to their energy scales is dependent on the kernel size of the decimator map C. The bulk space at fixed z will therefore not be evenly covered by the bulk variables, see figure 4.4. In order to define a holographic distance in a consistent way, we define the holograhic distance to be evaluated between different patches covered by a decimator filter. The distance between two filters A, B will then be the average geodesic distance between the bulk variables associated to it:

$$
\mathcal{D}_{\mathrm{holo}}(A, B) = \underset{\phi_\alpha \in A, \phi_\beta \in B}{\mathrm{avg}} \mathcal{D}_{\mathrm{geo}}(\phi_\alpha, \phi_\beta) \; . \tag{4.31}
$$

To probe the geometry we will calculate the angular distance within one layer and the radial distance across different z. The radial holographic distance has to be proportional to the number of layers r separating two decimators like

$$
\mathcal{D}_{\mathrm{holo}}^{rad}(A, B) \sim r \; . \tag{4.32}
$$

For the angular distance we expect for a hyperbolic space a logarithmic dependence on the difference in the coordinates as

$$
\mathcal{D}_{\mathrm{holo}}^{ang}(A, B) \sim \ln|r| \; , \tag{4.33}
$$

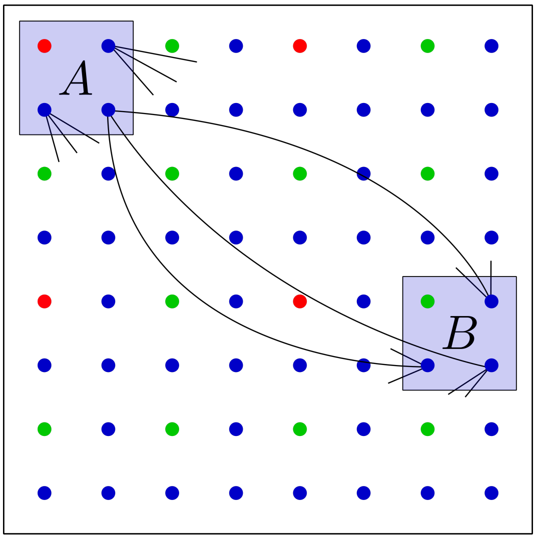


Figure 4.4: Distribution of bulk variables. Red dots represents IR modes, blue ones are UV modes and green dots are of intermediate length scale. The distance between two UV filters A,B is the average of the 9 paths connecting the associated blue dots.

with $r = \sqrt{x^2 + y^2}$, where A and B are separated by x and y decimators in the two lattice directions.

4.2.3 Loss function and saddle point

The probability distribution, that the network produces, can be calculated by a change of variables from the bulk distribution:

$$p_{\text{flow}}(\psi) = p_{\text{bulk}}(\phi) \left| \det \frac{\partial G(\phi)}{\partial \phi} \right|^{-1} \left| \det \frac{\partial G(\phi^\dagger)}{\partial \phi^\dagger} \right|^{-1} . \tag{4.34}$$

Since we transform $\mathrm{Im}(\phi)$ and $\mathrm{Re}(\phi)$ by the exact same map, the last to factors will give us the same value $\left|\det J_{G(\phi)}\right| \left|\det J_{G(\phi^\dagger)}\right| = \left|\det J_{G(\phi)}\right|^2$. For probability density distillation the reversed KL-divergence

$$D_{\text{KL}}(p_{\text{flow}}(\psi) \parallel q_{\text{CFT}}(\psi)) = \mathbb{E}_\phi \left[\ln p_{\text{bulk}}(\phi) - 2\ln|\det J_G| + S_{\text{CFT}}\right] + \ln Z \tag{4.35}$$

will be used, where the last term is constant and will be dropped in the implementation. The loss function will then be chosen as

$$\mathcal{L} = \mathbb{E}_\phi \left[\ln p_{\text{bulk}}(\phi) - 2\ln|\det J_G| + S_{\text{CFT}}[G(\phi)]\right] \tag{4.36}$$

The QFT will be evaluated on the configuration sampled from the network as $S_{\mathrm{CFT}} = S_{\mathrm{CFT}}[G(\phi)]$.

Let us interpret the above loss function in the context of holography. What does it mean to minimize the KL-divergence? Let's define the distribution for the gravity theory induced by the flow from the CFT density as

$$q_{\mathrm{grav}}(\phi, G) = q_{\mathrm{CFT}}(G(\phi))|\det J_G|^2 \ , \tag{4.37}$$

which will depend on the bulk field as well as on the parameters of the flow transformation contained in G. We can think of G as the background geometry, on which the field ϕ lives. As seen in the previous chapter, a training under the reversed KL-divergence is equivalent to training under the forward KL-divergence, where we try to match $q_{\mathrm{grav}}(\phi, G)$ to a prior distribution $p_{\mathrm{bulk}}(\phi)$. From equation (4.37) we also find an expression for the gravity action:

$$S_{\mathrm{grav}}[\phi, G] = S_{\mathrm{CFT}}[\psi] - 2\ln|\det J_G| \ . \tag{4.38}$$

Using this, the forward KL-divergence corresponding to the loss (4.36) is then given by

$$\begin{aligned} \mathcal{L} &= D_{\mathrm{KL}}(p_{\mathrm{bulk}}(\phi) \parallel q_{\mathrm{grav}}(\phi, G)) - \text{const.} \\ &= \mathbb{E}_\phi[-S_{\mathrm{bulk}}[\phi] + S_{\mathrm{grav}}[\phi, G]] \end{aligned} \tag{4.39}$$

This can be bounded from above as

$$\mathbb{E}_\phi[-S_{\mathrm{bulk}}[\phi] + S_{\mathrm{grav}}[\phi, G]] \leq \mathbb{E}_\phi[S_{\mathrm{grav}}(\phi, G)] \tag{4.40}$$

This shows, that minimizing the loss is equivalent to finding the saddle point approximation of the gravity theory. We can make an estimation for the marginal distribution $q_{\mathrm{grav}}(G)$ of the network's geometry encoded in G by

$$\begin{aligned} \frac{1}{Z_{\mathrm{grav}}} e^{-\mathbb{E}_\phi[S_{\mathrm{grav}}[\phi, G]]} &\leq \frac{1}{Z_{\mathrm{grav}}} \mathbb{E}_\phi\left[e^{-S_{\mathrm{grav}}[\phi, G]}\right] \\ &\leq \frac{1}{Z_{\mathrm{grav}}} \sum_\phi e^{-S_{\mathrm{grav}}[\phi, G]} \equiv q_{\mathrm{grav}}(G) \ , \end{aligned} \tag{4.41}$$

which holds by Jensen's inequality due to the convexity of the exponential. This implies that the marginalized probability of G is maximized under training, i.e. the geometry contained in G is classical. We therefore expect, that fluctuations are of small order and the bulk space has smooth behavior.

4.3 Bijective transformations

The complete EHM in generative direction consists of separate inverse RG-layers $G = \prod_k G_k$, which themselves contain a layer with decimators C_i and one with disentanglers E_i. In general those can be all chosen to be different, so $G_k = G_k(\{C_i\}_k, \{E_i\}_k)$. Since our physical system is translation invariant, it is reasonable to have the filters within a layer to share weights, such that G_k depends only on two filters C_k, E_k. One could go even further and impose scale invariance on the EHM, such that all RG-layers have the same transformation $G_{k+1} = G_k$.

C and E can in principle be chosen to be differentiable and invertible map as long as the logarithm of the Jacobian determinant (LDJ) and the inverse transformation can be calculated easily (like in RNVP networks). But we have seen that an EHM can already be achieved by a single matrix. So we expect our flow with two maps C, E to be expressive enough, if we choose them to be constructed with simple matrices W. Additionally we equip the net with a scaling layer s and an activation function a. The filters are then compositions of the form

$$\begin{aligned} C &= a \circ W_C \circ s \\ E &= a \circ W_E \circ s \ , \end{aligned} \tag{4.42}$$

where W will be varied by choosing from a number of options.

There is a another map L implemented before the first decimator layer in the direction of G, which is the reparametrization. It transforms a multivariate Gaussian with covariance $\Sigma = 1$ into a normal distribution, where the variables are correlated. Here, we also have freedom to constrain the weights in L.

Reparametrization To make the discretized bulk action $S_{\text{bulk}} = \phi^\dagger \Sigma^{-1} \phi$ reflect locality, we can restrict the form of the inverse covariance matrix by setting all entries to zero if two variables are not nearest neighbors within an RG layer or across different energy scales. The matrix will then have the form introduced in [18]

$$\Sigma^{-1} = \sum_{\langle \alpha, \beta \rangle} \kappa_{\alpha\beta} (|\alpha\rangle\langle\alpha| + |\beta\rangle\langle\beta| - |\alpha\rangle\langle\beta| - |\beta\rangle\langle\alpha|) + mI \ , \tag{4.43}$$

where I is the identity matrix, m is the mass of the field and $\kappa_{\alpha\beta} > 0$. Translation and rotational invariance at constant z can also be implemented if we use the same value for all bulk variables within the same layer. This reduces the number of parameters to only two per RG-step. But this covariance matrix will in general not be positive definite without additional conditions on $\kappa_{\alpha\beta}$ and m. Moreover, it is difficult to constraint Σ in such a way, that it stays positive-definite during training. Therefore, we opt for a different approach. Because the Cholesky decomposition into triangular matrices requires the original matrix to be positive-definite, we choose to build the covariance as $\Sigma = LL^\dagger$ and implement a simple linear layer L with the restriction to be lower triangular. The i.i.d. variables $\varphi_\alpha \sim \mathcal{N}_\mathbb{C}(0, 1)$ are then transformed to bulk states by

$$\phi = L\varphi \ . \tag{4.44}$$

This will ensure that Σ will always stay positive definite and ϕ follows a proper probability distribution.

Scaling In the direction of $G(\phi)$ the n incoming variables ϕ_α for one filter will be first scaled [18] by

$$s(\phi_\alpha) = e^{\lambda_\alpha}\phi_\alpha \tag{4.45}$$

with inverse

$$s^{-1}(\psi_x) = e^{-\lambda_x}\psi_x \ , \tag{4.46}$$

where the LDJ is simply given by the exponent λ.

Linear layer As our reference we choose a general $n \times n$ matrix. Since MERA and the EHM use unitary transformations, we will also implement them. At last we have an orthogonal map, which aims to reproduce the EHM introduced in [24]. The linear layer will then be a matrix from the following groups:

- $GL(n, \mathbb{C})$
- $SU(n)$ from a Cayley transform
- $SU(n)$ from a matrix exponential
- $O(4)$ from stacked $O(2)$ matrices (only for $n = 4$)

It is important to note, that the first three maps have complex weights and the last is restricted to real entries. The $O(4)$ transformation, which was used in [18], can be build from two block-diagonal matrices

$$Q(\omega_i, \omega_j) = \begin{pmatrix} \cos\omega_i & -\sin\omega_i & 0 & 0 \\ \sin\omega_i & \cos\omega_i & 0 & 0 \\ 0 & 0 & \cos\omega_j & -\sin\omega_j \\ 0 & 0 & \sin\omega_j & \cos\omega_j \end{pmatrix} \tag{4.47}$$

and

$$R(\omega_i, \omega_j) = \begin{pmatrix} \sin\omega_i & \cos\omega_i & 0 & 0 \\ \cos\omega_i & -\sin\omega_i & 0 & 0 \\ 0 & 0 & \sin\omega_j & \cos\omega_j \\ 0 & 0 & \cos\omega_j & -\sin\omega_j \end{pmatrix} \tag{4.48}$$

along with a matrix to permute two variables

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} , \tag{4.49}$$

where $R(\frac{\pi}{4}, \frac{\pi}{4})$ reproduces the unitary map from the EHM in its diagonal blocks.

The disentangler matrix is then defined as the composition

$$E_{O(4)} = Q(\omega_6, \omega_7) \circ P \circ Q(\omega_4, \omega_5) \circ Q(\omega_2, \omega_3) \circ P \circ Q(\omega_0, \omega_1) \tag{4.50}$$

and the decimator as

$$C_{O(4)} = R(\omega_6, \omega_7) \circ P \circ R(\omega_4, \omega_5) \circ Q(\omega_2, \omega_3) \circ P \circ Q(\omega_0, \omega_1) \ . \tag{4.51}$$

Activation The non-linearity contains no parameter fro training and is a hyperbolic function

$$a(\phi_\alpha) = \frac{\phi_\alpha}{|\phi_\alpha|} \operatorname{arsinh}|\phi_\alpha| \ , \tag{4.52}$$

since sinh(x) naturally appears in AdS spaces. The inverse and LDJ are only properly defined for $\phi_\alpha \neq 0$. In our implementation we choose to set the inverse and LDJ to zero for $\phi = 0$ and get

$$\log \left| \det \frac{\partial a}{\partial \phi_\alpha} \right| = -\frac{1}{2} \ln\bigl(1 + |\phi_\alpha|^2\bigr) \ . \tag{4.53}$$

In the coarse-graining direction G^{-1} the inverse activation is given by

$$a^{-1}(\psi_x) = \frac{\psi_x}{|\psi_x|} \sinh|\psi_x| \tag{4.54}$$

with

$$\log \left| \det \frac{\partial a^{-1}}{\partial \psi_x} \right| = \ln(\cosh|\psi_x|) \ . \tag{4.55}$$

4.4 Training and results

The flow was implemented using the ML library PyTorch along with the MERA structure from [18]. All networks were trained on a Nvidia GeForce GTX 1080 GPU. The $SU(n)$ matrices are build as parametrizations of a standard PyTorch `Linear(n,n)` layer. As standard hyper-parameters we choose:

CFT size	32×32
Kernel size	2×2
Batch size	1024
Stage I steps	60 000
Stage II steps	60 000

The learning rate for the linear maps is set to 10^{-4}, while for the scalings we decrease the rate by a factor of 10 every 20 000 steps from 10^{-2} to 10^{-3}. The weight in the reparametrization layer of the Gaussian is set to 10^{-3}.

4.4.1 Weight initialization

For the first stage of training we set the weights of the scaling layers to 0 to avoid an exploding loss function. Whenever possible, we initialized the linear layers to

$$W_E^{\text{init}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{4.56}$$

and

$$W_C^{\text{init}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{pmatrix} \ . \tag{4.57}$$

This corresponds to setting the matrices Q, R from the $O(4)$ to be: $Q = 1$ and R has the EHM transformation on the diagonal blocks. It seems to be a reasonable choice, since we know, that this map can produce an RG-scheme. For the linear layers we have to map the initialization back to the weights of the original linear layer by the reverse parametrization. For $SU(4)$ matrices from the Cayley map this is not possible, since W_E^{init} is not a solution of the inverse Cayley transform. The $SU(4)$ matrix from the exponential map cannot be initialized at all in general, because there is no matrix logarithm implemented in PyTorch. In these cases we will simply use the default random initial weights. All layers will have no bias term, so they are compatible with the $U(1)$ symmtery of the boundary theory.

For training stage II, the covariance matrix Σ or Cholesky matrix L should not be initiated as an identity matrix. Otherwise, the matrices stay diagonal during training. For our setup we choose to train with the approach of directly using the Cholesky matrix L, because the definition of Σ^{-1} with nearest-neighbour interactions does not stay positive-definite during our training and therefore gives an error message. The matrix L will be initialized after training stage I with a small perturbation on the off-diagonals as

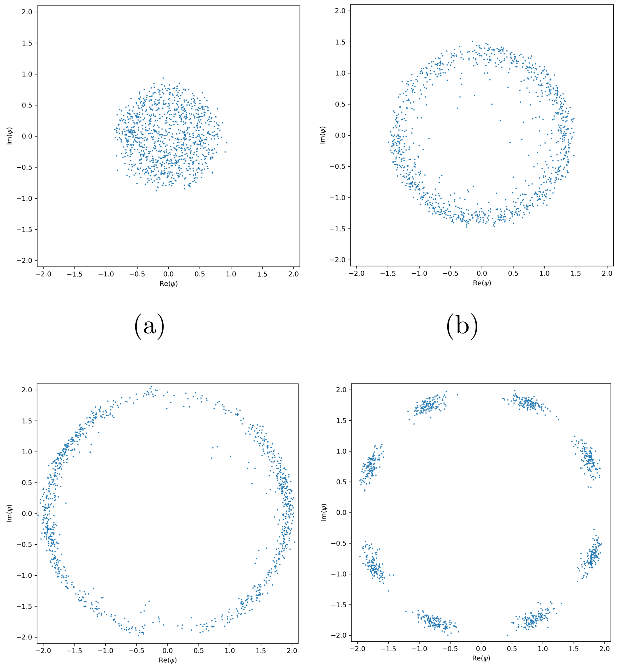
$$L^{\text{init}} = \begin{pmatrix} 1 & & \text{\huge 0} \\ & \ddots & \\ \text{\huge$\epsilon$} & & 1 \end{pmatrix} , \tag{4.58}$$

where the entries in $\boldsymbol{\epsilon}$ are drawn from a normal distribution $\mathcal{N}(0, 10^{-2})$.

4.4.2 Variation of filters

In the following we will denote the choice of filter for decimator C and disentangler E for example as (C, E) =(GL, GL), if we use linear layers for both. Let us look at the effect of the different choices on the training. We noticed that networks with exclusively $SU(4)$ or $O(4)$ matrices do not train. The distribution of ψ in the complex shows, that the field did not condensate into the minima. Using $GL(4)$ matrices for the C results in partial or slow training. Choosing $GL(4)$ for the distentanglers and $SU(4)$ or $O(4)$ for the decimators enables the system to fall into a ground state. For the most general transformation with only $GL(4)$ matrices we frequently observe a collapse into a mode with periodicity, see figure 4.5.

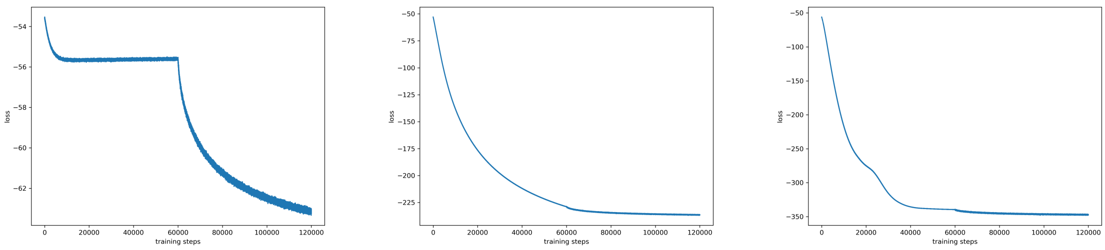
The drop of the loss function reflects how much the field has fallen into the circle, see figure 4.6. Since we train C, E during stage I, the loss function is affected in that region. The higher the energy of the field is, the less the loss is decreased before step 60 000.



(a) (b) (c) (d)

Figure 4.5: Distribution of ψ in the complex plane with transformation (C, E)
(a) SU, SU (b) GL, SU (c) SU, GL (d) GL, GL

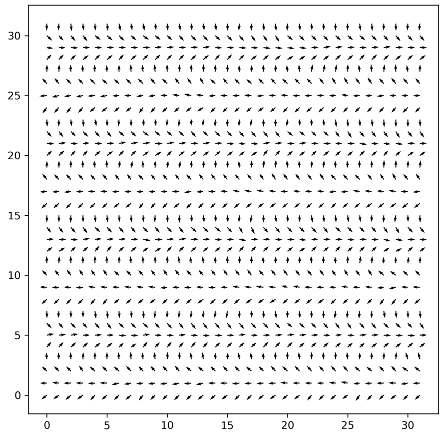
After the inital steep drop we see, that it is only slowly decreasing as usual in neural network training. The network can be further trained, when we enter stage II training at 60 000 steps. The covariance matrix of the bulk distribution seems to be capturing some correlation features, which could not be resolved by training the network weights alone. However, this does not necessary imply that these correlations are meaningful.



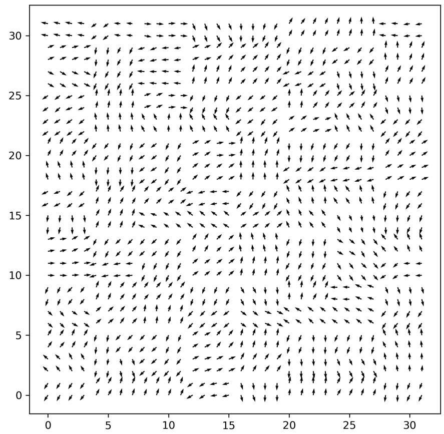
(a) (b) (c)

Figure 4.6: Loss for transformations (C, E)
(a) SU, SU (b) GL, SU (c) SU, GL

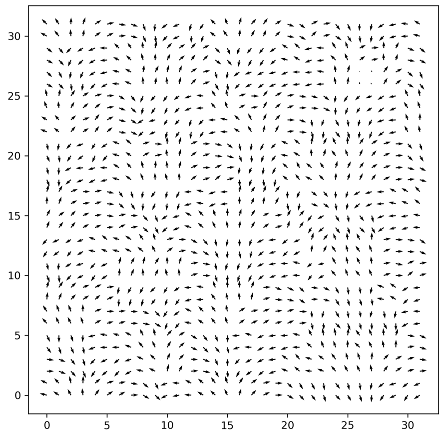
After training we observe different configuration types depending on the choice of transformations. We show the cases, where the disentangler E has a $GL(4)$ matrix. Using only linear maps, the network should be expressive enough to capture any of the other behavior. But it seems, that instead it experiences a mode collapse, which leads to a periodic structure (most apparent in the complex plane). For the unitary maps, we see some a pattern, that looks like the formation of magnetic domains. In the orthogonal case, the configuration is starting to acquire vortices.



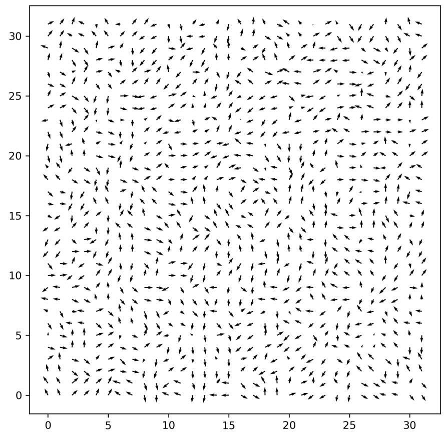
(a)



(b)



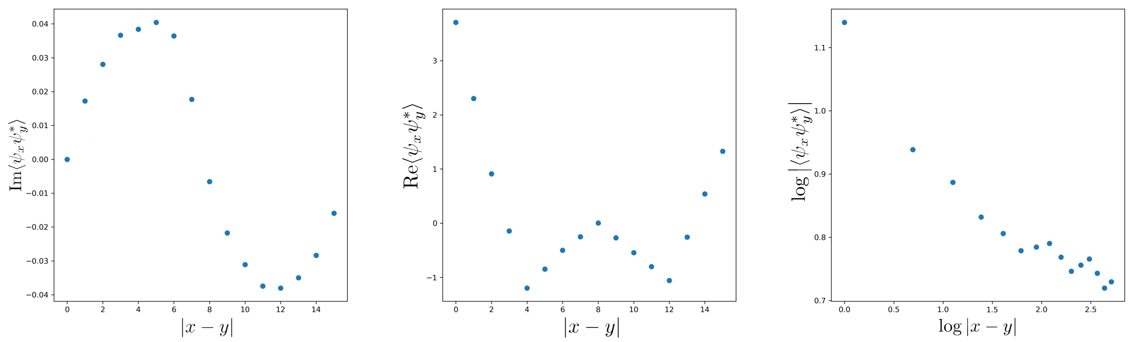
(c)



(d)

Figure 4.7: Loss for transformations (C, E)
(a) GL, GL (b) SU Cayley, GL (c) SU Exp, GL (d) O, GL

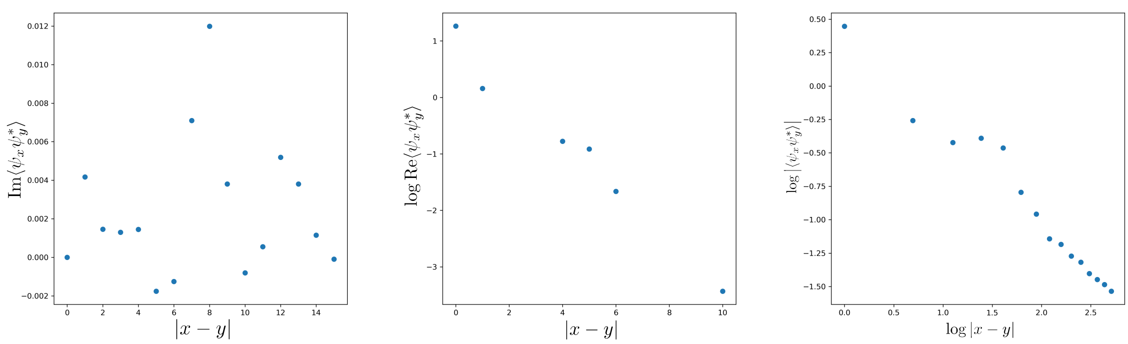
To see which configuration closest to an actual physical state, we look at the two-point functions. For the choice (SU, GL) the imaginary part is close to zero as expected, but has a periodic form. This implies, that the configuration is unphysical. A look at the real part reveals again a peculiar behavior. For the expected power-law the log-log plot should give a linear graph. This is almost the case for the absolute value $\left|\langle\psi_x\psi_y^*\rangle\right|$, see figure 4.8.



(a) (b) (c)

Figure 4.8: 2-point function for (SU, GL)
(a) $\mathrm{Im}\langle\psi_x\psi_y^*\rangle$ (b) $\mathrm{Re}\langle\psi_x\psi_y^*\rangle$ (c) $\left|\langle\psi_x\psi_y^*\rangle\right|$ log-log

For the orthogonal map the configurations are more physical. The imaginary two-point function is distributed around zero and shows no particular behavior that could immediately imply some functional dependence on the distance. The real part however shows an exponential decay, while the absolute value of the correlator shows the power-law, see figure 4.9.



(a) (b) (c)

Figure 4.9: 2-point function for (O, GL)
(a) $\mathrm{Im}\langle\psi_x\psi_y^*\rangle$ (b) $\mathrm{Re}\langle\psi_x\psi_y^*\rangle$ in log-lin(c) $\left|\langle\psi_x\psi_y^*\rangle\right|$ in log-log

It seems, that our network has not captured the physics entirely. As a check we look at the holographic distances for the $(O(4), GL)$ network. The plots do not show the expected behavior, see figure 4.10. If the flow would have a hyperbolic geometry, we would see a linear distribution in both graphs.

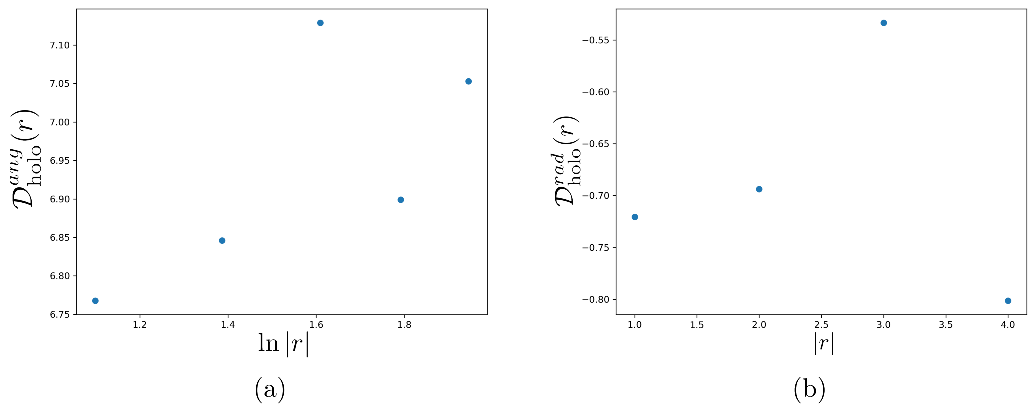
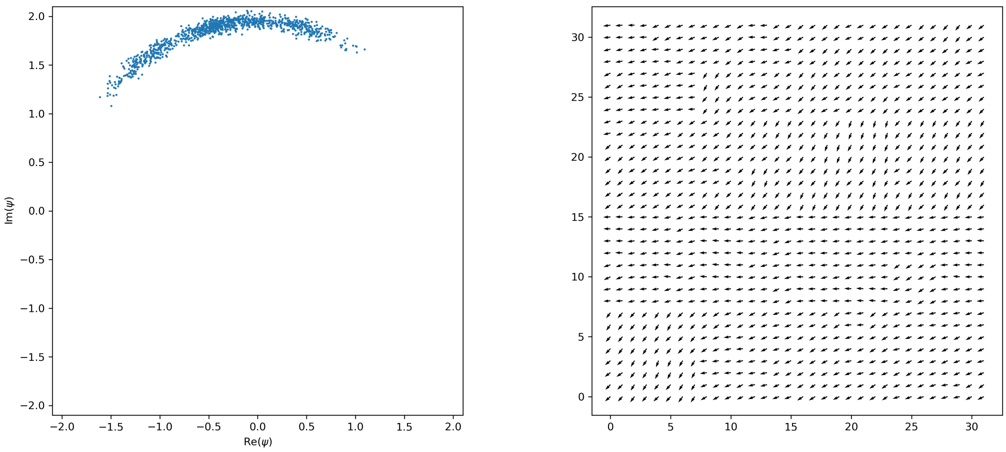


Figure 4.10: Holographic distances: (a) angular (b) radial

Longer training Because we expect the loss function to continue to drop during the optimization, we will train a model for the double amount of training steps. We use the (SU Cayley, GL) model and keep all parameters the same, expect for the steps, which are now 120 000 per training stage. We observe, that the network starts randomly select certain field directions in the complex plane. That means, it has acquired a quasi-long-range order. The $U(1)$ symmtery is restored across multiple samples, see figure 4.11.

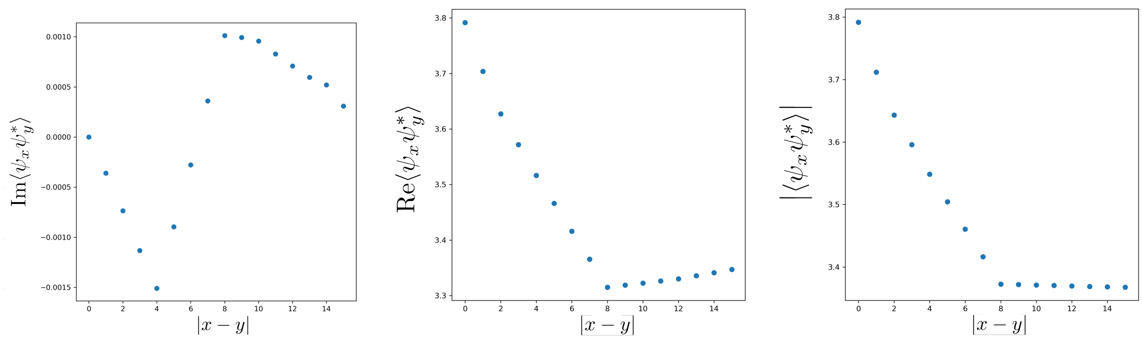
The two-point function shows confirms the theoretical results. The imaginary part is approximately zero, while the real part and absolute value have the same behavior. However, they do not show an polynomial decay but a linear dependence on $|x - y|$ instead, see figure 4.12. This is due to the formation of domain walls, which separate regions with constant direction. We expect the domain sizes to grow, when we continue the training process.

The holographic distances show basically the some behavior as before with no clear dependence. Apparently the flow network was not able to find stable configuration for the bulk field. The network might be stuck in some local minimum, which produces almost physical configurations but without a hyperbolic bulk structure. Or in other words, the geometry seems to not have reached its classical limit. This is supported by the fact, that during stage II training the loss function is heavily fluctuating, see figure 4.13.



(a) (b)

Figure 4.11: (SU Cayley, GL) model: (a) distribution (b) configuration



(a) (b) (c)

Figure 4.12: (SU Cayley, GL) model trained for 240k steps:
(a) $\mathrm{Im}\langle\psi_x\psi_y^*\rangle$ (b) $\mathrm{Re}\langle\psi_x\psi_y^*\rangle$ (c) $\left|\langle\psi_x\psi_y^*\rangle\right|$

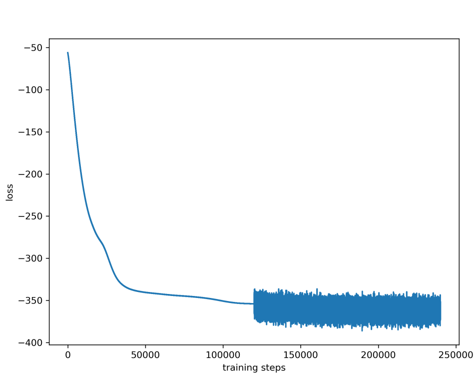


Figure 4.13: Loss for (SU Cayley, GL) model trained for 240k steps.

Summary The different implementations for constructing unitary maps affect the network only slightly. This is to be expected as they produce almost the same set of matrices. The training with only linear layer is very unstable. While it can reach configurations like the ones with unitary maps, the network experiences frequent mode collapses. The orthogonal layers are the most restricted and therefore also take the longest to train. We sum up the investigation in table 4.1.

decimator	disentangler	observation
GL	GL	periodic structure
SU	GL	domains
O	GL	vortices
GL	any	slow convergence
SU/O	SU/O	not training

Table 4.1: Effects of different filter choices

4.5 Model extensions

To get further investigate the flow network, we will test two small extension, which are motivated from physics. The first one is the inclusion of the source term in the action, which has the role of a coupling to an external (magnetic) field. The other is to fully implement conformal symmtery into the model.

External field The bias term should be set to 0, because otherwise it would break the rotation symmetry of the action. Training with a bias leads to mode collapse, where the system picks a random direction for the fields during training and is then stuck at it. However, if we want to include an external field ϕ_0, a non-zero bias term is necessary to capture this effect, see figure 4.14. The network picks up the direction defined by the external source and stays in roughly that configuration over multiple samples.

Although the ϕ_0 could be seen as a source, it has no connection to the bulk field. It is simply an external parameter and does not correspond to a boundary value of the bulk field ϕ.

Scale invariance Because the boundary CFT is theoretically in critical phase, we could impose scale invariance on the network. As an example we work with the choice (SU Cayley, GL) and include scale invariance, i.e. the same filters are used across all layers.

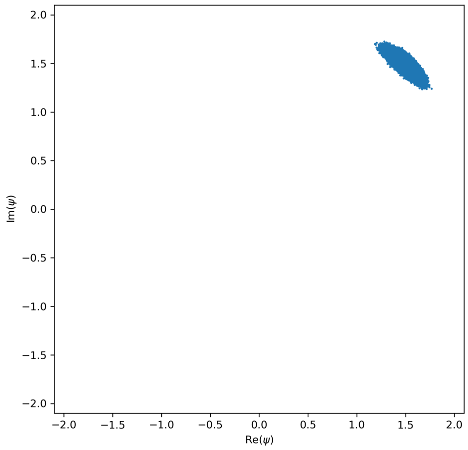
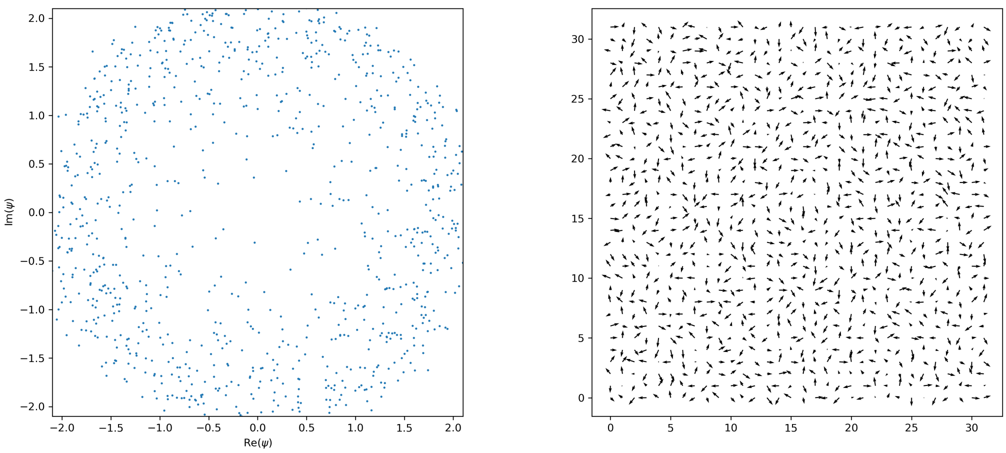


Figure 4.14: 1024 samples with external field $\phi_0 = 100 + 100\,\mathrm{i}$.

This reduces drastically the number of parameters and results in faster training. But it seems, that the flow also struggles to bring the system into the ground state. After training the same number of steps as the networks before, we see that the field has started to condensate into the minima, but much slower than the flow without scale invariance, see figure 4.15. The training was done with 120 000 steps in total.



(a) (b)

Figure 4.15: Network with (SU Cayley, GL) and scale invariance
(a) field configuration (b) distribution in complex plane

5 Outlook

The holographic network has some convenient features, that can be used for different task. The AdS/CFT correspondence in general enables us to study theories in a strong coupling regime, because the dual theory will then be weakly coupled. Although the neural network goes into the classical limit of the bulk under training, the EHM is in general valid at the quantum level. The computation of observables like n-point function can performed by using the bulk partition function.

5.1 Sampling

We can use the network to boost the efficiency of Monte Carlo sampling of lattice QCD configurations. This has been proposed in [1] and [2]. We can take the Metropolis-Hasting algorithm and use the probability density p_{flow} provided by the network as a proposal distribution. In a specific setup we want to the sample distribution to converge to the CFT distribution p_{CFT}. So given a Markov chain, the proposal ψ' sampled from the holographic network is accepted with probability

$$\mathcal{A}(\psi^{(i-1)}, \psi') = \min\left(1, \frac{p_{\text{flow}}(\psi^{(i-1)})p_{\text{CFT}}(\psi')}{p_{\text{CFT}}(\psi^{(i-1)})p_{\text{flow}}(\psi')}\right) , \tag{5.1}$$

where we set $\psi^{(i)} = \psi'$ in the case of acceptance. It has been demonstrated [21], that the efficiency of the sampling is much better when using the flow network distribution. This is of course not that surprising given that p_{flow} was build to be a tractable approximation of the desired probability density.

5.2 Symmteries

If a system has a defining or fundamental symmetry, then the dynamics and all future system states will respect that underlying symmetry. It makes sense to implement these structures into networks, because models which heavily violate the symmetries should not be able to capture the information about the system anyway. In case of emergent symmetries one should be careful, since these can be present in our data sets but are not guaranteed to be. Networks which respect these symmetries are called equivariant networks. A general introduction and overview of equivariant networks is [9]. In our case, we have a $U(1)$ symmetry together with a reflection symmetry $\mathbb{Z}_2$.

For matrix groups, there exist a general approach to constraint the weight matrix [12]. A linear map represented by a matrix M is said to be equivariant under a group G if it satisfies In general we can say that the weights in M transform under some representation $\rho(g)$:

$$\rho(g)M = M \ , \tag{5.2}$$

where v_M is the flattened matrix M. This can be written as an action of the generators on the object M by

$$\mathrm{d}\rho(\mathfrak{g}_i)M = 0 \ . \tag{5.3}$$

Solving the constraint induced by the Lie-algebra is sufficient to constraint M to be equivariant under the group G. In this way we could implement additional symmetries besides $U(1)$ rotation. If we implement theories with multiple fields, stored in the channel dimension, symmteries like SUSY can be incorporated as a constraint on the disentangler or decimator transformations.

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Appendix

The code can be found on Gitlab under:
`https://gitlab.physik.uni-muenchen.de/Sven.Krippendorf/geometric-dl-holography`

To train a model one can run the file `main.py` with pyton. Unless parameters and options are specified by flag, the default values in `args.py` are used.

As an example we choose:

- unitary map (Cayley) as decimator
- linear map as disentangler
- train on GPU

For this we run the following command:

```
python main.py --decimator su_cayley --disentangler linear --cuda 0
```